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An Integrative Framework for Sex-Specific Microglial Morphometry and Phenotyping in Alzheimer's Disease

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Abstract

Alzheimer's disease (AD) exhibits pronounced sex differences in prevalence and progression, yet the interaction between biological sex, microglial morphology, and cognitive outcomes remains poorly characterized. Microglia play a central role in synaptic regulation and neuroinflammation, and changes in their morphology are associated with shifts in functional state. We present a computational analysis of hippocampal microglial morphology and behavioral performance in female and male 5xFAD (a transgenic Alzheimer's model) mice alongside wild-type controls. Using digitally reconstructed microglia from NeuroMorpho.org, we analyzed nearly 4,000 cells across multiple morphometric features. Cognitive performance measures from the open-access rodent database MouseBytes were also examined. Morphological comparisons revealed widespread alterations between AD and control groups, with soma size and branching metrics accounting for the largest sources of variance. Sex-stratified comparisons with Bonferroni correction confirmed that most differences remained significant. Supervised machine learning models distinguished AD from control microglia with up to 96% accuracy, with feature-attribution analyses highlighting soma- and branch-related features as highly informative. Sex stratification revealed that AD-associated morphological simplification was more pronounced in females across nearly all metrics, with larger effect sizes compared to males. Behavioral analyses of five-choice serial reaction time task (5CSRTT) performance revealed distinct sex-specific cognitive impairments; males demonstrated broader and more statistically robust impairments compared to females in perseverative correct responses, correct response latency, reward collection latency, and average omission rate. These findings demonstrate that AD-related microglial and cognitive alterations are strongly sex-dependent, highlighting the value of integrative, data-driven approaches for studying neuroimmune mechanisms in Alzheimer's disease.

Keywords: Alzheimer's disease, microglia, morphology, sex differences, machine learning, cognition

1. Introduction

Alzheimer's disease (AD) is a progressive form of dementia that affects over 7 million individuals in the United States (Rajan et al. 2021). With an aging population, this number is projected to increase significantly in the coming decades, emphasizing the urgent need to investigate factors that contribute to disease onset and progression. Advancing our understanding of AD aligns directly with United Nations Sustainable Development Goal 3.4 (Zhang et al. 2025), which calls for reducing premature mortality from non-communicable diseases and promoting neurological health worldwide. Considerable progress has also been made in AD diagnosis, including the identification of candidate biomarkers (Yu et al. 2021), development of neuroinformatics resources (Li and Liang 2022), and advances in deep learning-based diagnostic tools (Raza et al. 2025).

While amyloid-beta plaque accumulation and tau protein aggregation into neurofibrillary tangles are well-established characteristics of AD, the precise cellular mechanisms underlying neuronal loss remain incompletely understood (Sarlus and Heneka 2017). Quantitative, cell-level analyses are increasingly needed to link cellular alterations to disease-related functional outcomes.

Recent evidence indicates that microglial cells, the primary immune cells of the brain, play a central role in disease development. Microglia in the hippocampus, a region critical for learning and memory, are highly active and attempt to clear amyloid-beta. However, aging and genetic risk factors can induce a shift to a disease-associated or "toxic" state that compromises microglial function, facilitates plaque accumulation, and promotes chronic neuroinflammation through cytokine release that further damages neuronal integrity (Wu et al. 2024; Valiukas et al. 2025; Sarlus and Heneka 2017). These functional transitions are accompanied by characteristic changes in microglial morphology, which range from the highly ramified, homeostatic forms that survey large areas of brain tissue, to simplified reactive or amoeboid morphotypes with retracted processes and enlarged somas, making morphometric analysis a useful approach for systematically assessing microglial state across disease conditions (Reddaway et al. 2023).

Emerging evidence further suggests that sex modulates microglial activity and AD pathophysiology. Transcriptomic studies have identified female-enriched, disease-associated microglial subtypes characterized by upregulated inflammatory and phagocytic gene expression, linked to accelerated amyloid pathology and greater AD severity (Wu et al. 2024; Coales et al. 2022), while systems-level analyses indicate sex-specific molecular networks and key drivers of AD (Guo et al. 2023). Imaging and neuropathology data in humans support these findings, showing that microglial activation disproportionately affects females (Casaletto et al. 2022; Reed and Keller-Norrell 2023). Together, these studies indicate that sex is a key modulator of microglial behavior in AD, yet its impact on microglial structure and associated cognitive outcomes remains incompletely quantified (Vila-Castelar et al. 2024).

The present study aims to fill this gap by systematically characterizing hippocampal microglial morphology in healthy and AD-model mice, addressing the following three research questions:

> **RQ1. What morphometric features distinguish hippocampal microglia in 5xFAD mice from wild-type controls?**
>
> **RQ2. Do AD-associated microglial morphological changes differ in magnitude or pattern between male and female 5xFAD mice?**
>
> **RQ3. Do 5xFAD-associated cognitive impairments, as measured by 5CSRTT performance, show sex-specific profiles, and how these compare with the morphological findings?**

2. Methods

This study employed a computational pipeline integrating large-scale microglial morphometry, statistical analysis, and machine learning to characterize Alzheimer's disease (AD)-related and sex-dependent differences in microglial structure and associated cognitive outcomes, including attention, impulse control, and cognitive flexibility as measured by the 5-choice serial reaction time task (5CSRTT) (Robbins 2002).

Microglia were selected as the primary unit of analysis because their morphology directly reflects functional state; shifts from highly branched, homeostatic forms to simplified, activated morphologies are among the earliest and most consistent cellular changes observed in AD (Sarlus and Heneka 2017; Davies et al. 2017; Baron et al. 2014). Digital reconstructions of microglial morphology, three-dimensional representations of cellular structure encoded as node coordinates and branching relationships, enable quantitative morphometric analysis at a scale not achievable through manual methods (Kozlowski and Weimer 2012; Megjhani et al. 2015).

We first describe the acquisition and preprocessing of microglial reconstructions, followed by exploratory and inferential statistical analyses. We then detail supervised and unsupervised learning approaches for classification and dimensionality reduction and finally outline the integration of behavioral performance data. An overview of the analytical workflow is provided in Fig. 1.

Data were obtained from two open-access repositories: NeuroMorpho.org (microglial SWC reconstructions) and MouseBytes (5CSRTT behavioral records). RQ1 was addressed through an exploratory data analysis (EDA) pipeline comprising data acquisition and preprocessing, exploratory visualization and statistical analysis, dimensionality reduction and clustering, and supervised classification; RQ2 and RQ3 were each addressed through analogous sex-stratified acquisition, preprocessing, and statistical analysis of morphometric and behavioral data, respectively. Findings from RQ2 and RQ3 were then integrated to characterize sex-dependent microglial and cognitive alterations in Alzheimer's disease.

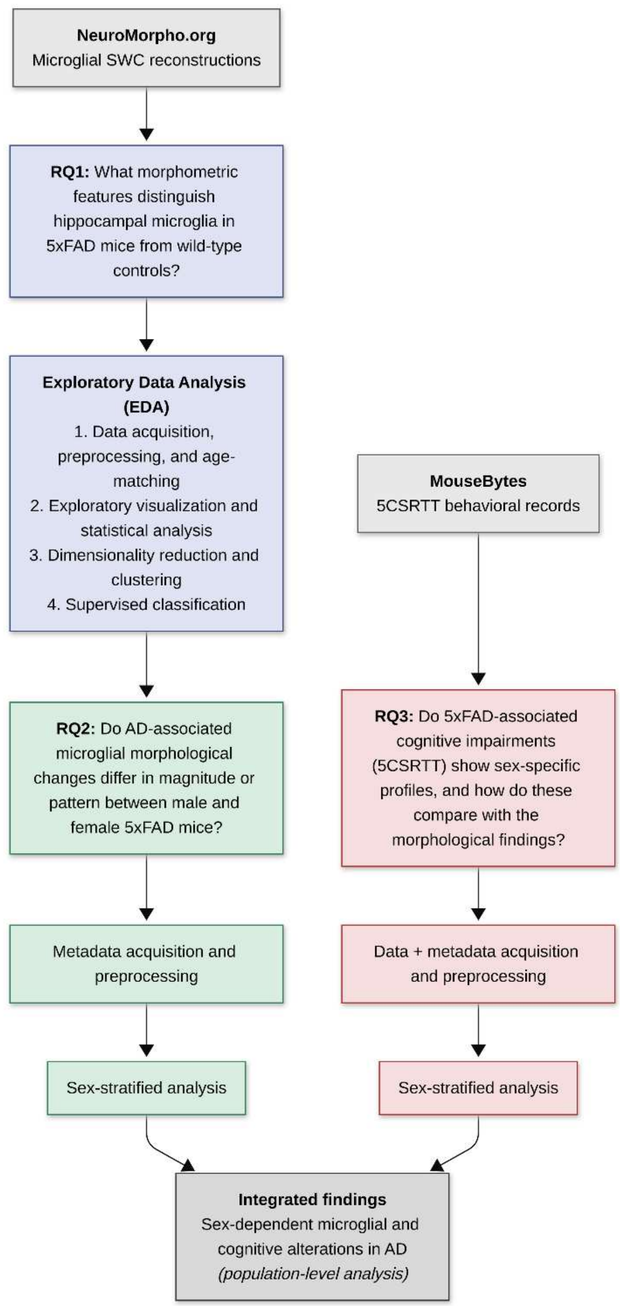


Fig. 1 Computational analysis pipeline organized by research question

2.1. Microglial datasets and morphometric features

To investigate AD-related and sex-dependent morphological changes, we analyzed microglial cells from healthy control mice and from 5xFAD mice, a transgenic model of Alzheimer's disease that coexpresses five familial AD mutations in amyloid precursor protein and presenilin-1, resulting in accelerated amyloid deposition beginning as early as 2 months, progressive neuronal loss, and memory impairment that collectively recapitulate key features of human AD pathology (Oakley et al. 2006). Digitally reconstructed microglia were obtained from NeuroMorpho.Org, an open-access repository of neural and glial cell morphologies (Ascoli et al. 2007). Cellular reconstructions were provided as SWC files (Online Resource 1; SWC Specification Governing Board 2023), which encode three-dimensional cellular structure using node coordinates, radii, and parent-child relationships (Online Resource 2). Representative examples of reconstructed microglial morphologies from control and 5xFAD mice are shown in Online Resource 3.

For the 5xFAD cohort, all available hippocampal microglia were included. To mitigate class imbalance, a matched number of control microglia were randomly sampled (random seed = 42). Morphometric processing and feature extraction were performed using the NeuroM Python library (v4.0.4; Blue Brain Project 2024). Extracted features included soma volume, soma surface area, soma radius, and seven additional morphometric measures selected based on prior computational studies linking microglial morphology to functional and activation states (Online Resource 4; Kozlowski and Weimer 2012; Reddaway et al. 2023; Verdonk et al. 2016; Megjhani et al. 2015; Colombo et al. 2022). Because the number of bifurcations is deterministically related to the number of sections in a bifurcating tree, these two metrics are near-perfectly collinear (see Results) and should be interpreted as a single branching-complexity axis rather than as independent features. Crucially, per-cell demographic metadata, specifically categorical age classifications and sex annotations, were programmatically extracted from NeuroMorpho.org alongside the structural files. These metadata fields were utilized to apply a strict, global age-matching filter at the absolute baseline of the pipeline, ensuring that all downstream exploratory, statistical, and machine learning analyses were restricted exclusively to a highly controlled adult cohort.

2.2. Exploratory Data Analysis (RQ1)

2.2.1. Data acquisition and preprocessing

The initial dataset comprised 6,035 microglial cell reconstructions. Twenty-five control reconstructions exhibiting soma extraction errors were excluded, resulting in 6,010 total microglia, all of which were successfully merged with their respective metadata files. This unified dataset was then filtered using the co-extracted metadata to isolate an adult-only cohort, controlling for age-related confounds across all subsequent analyses. Animals with ambiguous or non-adult classifications, along with those lacking sex metadata, were removed. This age-matching and demographic filtering yielded a baseline adult cohort of 2,642 control and 1,176 AD microglia (3,818 cells total). Outliers were defined as values exceeding three standard deviations from the mean for any morphometric feature.

Two dataset versions derived from this adult cohort were used, selected based on each analytical approach's sensitivity to outliers and class imbalance. For exploratory visualization and statistical comparisons, a filtered, outlier-removed dataset reduced the influence of extreme values (2,416 control and 1,169 AD microglia; 3,585 total). For supervised classification, an outlier-retaining balanced dataset was used, with the baseline Control group down sampled to match the AD count (1,176 per class; 2,352 cells total; random seed = 42) to preserve raw morphological variance while mitigating class imbalance. Where class balance was additionally required for dimensionality reduction and clustering, outliers were removed after balancing, yielding 1,043 control and 1,167 AD cells (2,210 total).

2.2.2. Exploratory visualization and statistical analysis

Exploratory analyses were conducted to characterize morphometric differences between groups and assess feature distributions. To reduce the influence of extreme values on visualization and summary statistics, these analyses were performed on the filtered dataset. Univariate comparisons were visualized using boxplots stratified by disease status (showfliers=False, unbalanced classes), while multivariate relationships were examined using pairwise scatter plots and correlation heatmaps (balanced and unbalanced classes). Group differences were evaluated using Welch's two-sample t-tests, a statistical approach for comparing means between two independent groups that does not assume equal variances, on both the balanced and unbalanced filtered datasets. This framework was selected to assess morphometric differences between AD and control microglia under potential heteroscedasticity. Morphological variance within and between groups was also examined to assess heterogeneity in microglial structure, as high intragroup variance may reflect functionally distinct microglial subpopulations rather than a uniform response to AD pathology.

To account for the increased risk of Type I errors (false positives) associated with testing multiple morphological features simultaneously, family-wise error rate (FWER) control was implemented. For each cohort analysis, independent-samples Welch's t-tests were performed across the ten distinct microglial morphometric metrics, and the resulting raw p-values were adjusted post-hoc using the Bonferroni correction method, in which each raw p-value was multiplied by the total number of comparisons within that test family (n = 10). Statistical significance across all univariate comparisons was defined as an adjusted p-value (p_{adj}) of less than 0.05.

2.2.3. Dimensionality reduction and clustering

Principal component analysis (PCA), a dimensionality reduction technique that transforms correlated variables into a smaller set of uncorrelated components ordered by the variance they explain, was applied to the filtered, class-balanced subset (2,210 total cells) to visualize dominant axes of morphometric variance, with the first two components used for visualization and identification of the strongest contributing features. K-means clustering, an unsupervised algorithm that partitions data into k groups by iteratively assigning each point to its nearest centroid until convergence, was performed on both PCA-informed and full morphometric feature sets, using Euclidean distance as the similarity metric. Clustering quality was evaluated using silhouette scores (range -1 to 1) and by visual comparison to the PCA.

2.2.4. Supervised classification

Supervised machine learning refers to the training of algorithms on labeled data to learn patterns that generalize to new, unseen samples. Here, supervised learning models were trained to classify microglia as AD or control based on extracted morphometric features. Unlike the clustering pipeline, this classification framework utilized the class-balanced, outlier-retaining dataset (2,352 total cells), which was standardized using Z-score normalization (*StandardScaler*) prior to model training.

Three classifiers were implemented using the scikit-learn library in Python (Pedregosa et al. 2011). Logistic regression, used as a linear baseline, estimates the probability of class membership as a weighted combination of input features (*max_iter* = 1000). Random Forest, an ensemble method that aggregates predictions across multiple decision trees (*n_estimators* = 100) trained on random feature subsets, was used to capture nonlinear relationships. Gradient Boosting, a sequential ensemble approach in which each tree corrects the residual errors of the previous one (*n_estimators* = 100, *learning_rate* = 0.1), was also applied for its strong predictive performance on tabular data. A global random seed of 42 was set for all algorithms to ensure exact reproducibility.

Model performance was assessed using a stratified five-fold cross-validation strategy (*StratifiedKFold*), with stratification ensuring that class proportions were strictly preserved across folds. Evaluation metrics included accuracy, confusion matrices, and receiver operating characteristic (ROC) curves.

To enhance interpretability and link model predictions to biological features, feature importance was assessed using both model-specific measures and SHapley Additive exPlanations (SHAP), a model-agnostic feature attribution framework applicable to any classifier regardless of its internal structure (Lundberg and Lee 2017), to quantify the contribution of each morphometric feature to model predictions.

2.3. Sex-stratified metadata analyses (RQ2)

Sex annotations were programmatically extracted using the NeuroMorpho.Org API. Of the 6,010 merged reconstructions, 2,192 were removed as non-adult or lacking unambiguous sex annotation (including 317 'Male/Female' entries), yielding 3,818 adult cells; subsequent outlier removal (233 cells) produced the 3,585-cell sex-stratified cohort (1,349 female and 1,067 male control; 653 female and 516 male 5xFAD). Main effects and potential interaction terms were evaluated using independent Welch's t-tests and two-way analysis of variance (ANOVA). To account for multiple comparisons across the ten morphometric features, p-values were adjusted post-hoc using the Bonferroni method. For the sex-stratified pairwise Welch's t-tests, alpha adjustments were applied across a pooled family of n = 20 comparisons (10 morphometric features × 2 sexes). For the two-way ANOVA, multiplicity adjustments were applied independently per individual analytical factor (disease group, biological sex, and their interaction term) across a localized family of n = 10 comparisons (10 morphometric features per factor). Statistical significance across all metrics was defined as an adjusted p-value (p_{adj}) of less than 0.05.

2.4. Behavioral data incorporation (RQ3)

To examine links between microglial morphology and cognitive performance, behavioral data were obtained from MouseBytes, an open-access high-throughput rodent touchscreen platform (Beraldo et al. 2019). Specifically, metrics from the 5CSRTT were extracted and analyzed across nine behavioral domains, each capturing a distinct aspect of attention and cognitive control (Online Resource 5).

The 5CSRTT is a validated behavioral test that assesses visual attention and motor impulsivity by requiring the subject to monitor five distinct apertures for a brief light stimulus. Successful localization and nose-poking of the illuminated aperture results in a reward, while premature or incorrect responses are recorded as measures of impaired inhibitory control or distractibility.

To account for multiple comparisons across the nine behavioral features, p-values were adjusted post-hoc per analytical factor using a Bonferroni adjustment (n = 18 comparisons per family: 9 behavioral domains × 2 sexes), with statistical significance defined as an adjusted p-value (padj) of less than 0.05. Independent group-level alterations and sex-specific trends within these behavioral domains were analyzed in parallel with tissue-derived morphometric profiles to contextualize structural microglial phenotypes alongside cognitive outcomes.

Behavioral data were filtered using the same outlier removal criteria applied to morphometric features. The final dataset consisted of 2,317 control (C57BL/6J: n = 1,595; B6129SF2/J: n = 722) and 992 5xFAD samples.

2.5. Reproducibility and experimental setup

All analyses were performed in a cloud-based Python environment (Google Colaboratory) using standard CPU and GPU resources. Random sampling and cross-validation procedures were performed with fixed random seeds to ensure reproducibility. All code and datasets are publicly available at: https://github.com/AvniSriram/Sex-Specific-AD-Microglia.git.

3. Results

3.1. Morphological complexity (RQ1)

Boxplots were examined to assess differences between AD (5xFAD) and control microglia across all extracted morphological metrics, as seen in Fig. 2. Visual inspection suggested reduced morphological complexity in 5xFAD microglia across most metrics. Univariate group differences were evaluated using Welch's two-sample t-tests restricted to the age-matched adult-only cohort. Before applying multiple-comparison adjustments, all ten morphometric metrics demonstrated statistically significant differences between groups (praw < 0.05). However, following post-hoc Bonferroni correction ($n = 10$), the number of neurites, mean segment length, and mean local bifurcation angle did not maintain statistical significance (padj ≥ 0.05). All remaining microglial metrics successfully retained their significance post-correction.

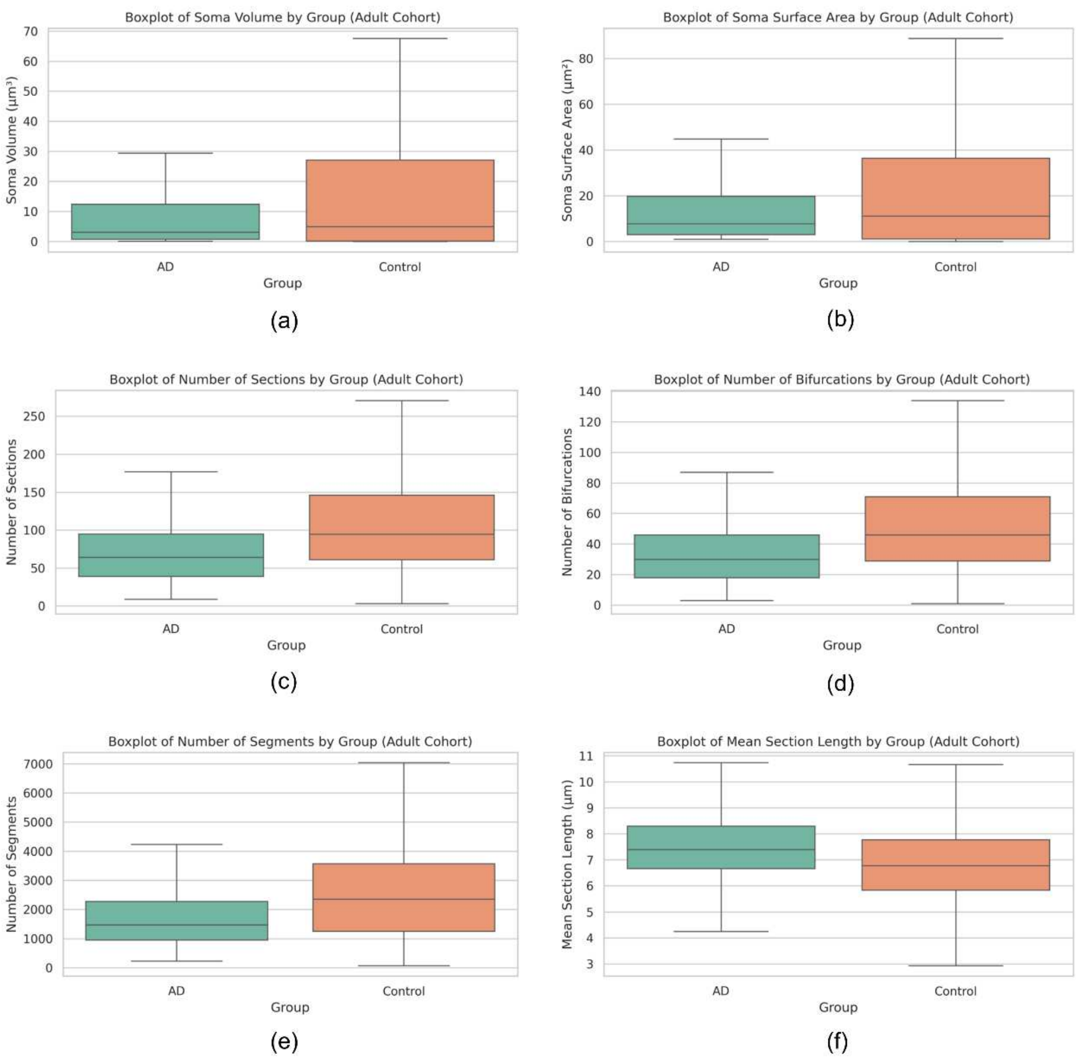


Fig. 2 Representative box plots showing quantitative differences in most significant morphometrics between 5xFAD (AD) and wild-type control microglia. Metrics include (a) soma volume, (b) soma surface area, (c) number of sections, (d) number of bifurcations, (e) number of segments, and (f) mean section length

The reduction in soma volume and surface area observed in 5xFAD microglia (panels a, b) is consistent with a transition away from the highly ramified homeostatic phenotype, in which larger soma support extensive process elaboration, toward morphologically simplified states associated with microglial activation or dystrophy (Sarlus and Heneka 2017). The reduced branching complexity reflected in lower section and bifurcation counts (panels c, d) further supports a shift away from surveillance-competent morphology. Similarly, the decrease in number of segments (panel e) reflects a global reduction in process elaboration in 5xFAD microglia. Notably, mean section length was modestly elevated in 5xFAD microglia (panel f) despite reduced overall branching, suggesting that surviving processes may be geometrically elongated as local branch points are lost, which may indicate partial process retraction rather than complete arbor elimination.

Pairwise correlations between morphometric features were calculated and visualized as both pairwise scatter matrices (Online Resource 6) and Pearson correlation heatmaps (Fig. 3) for the balanced dataset.

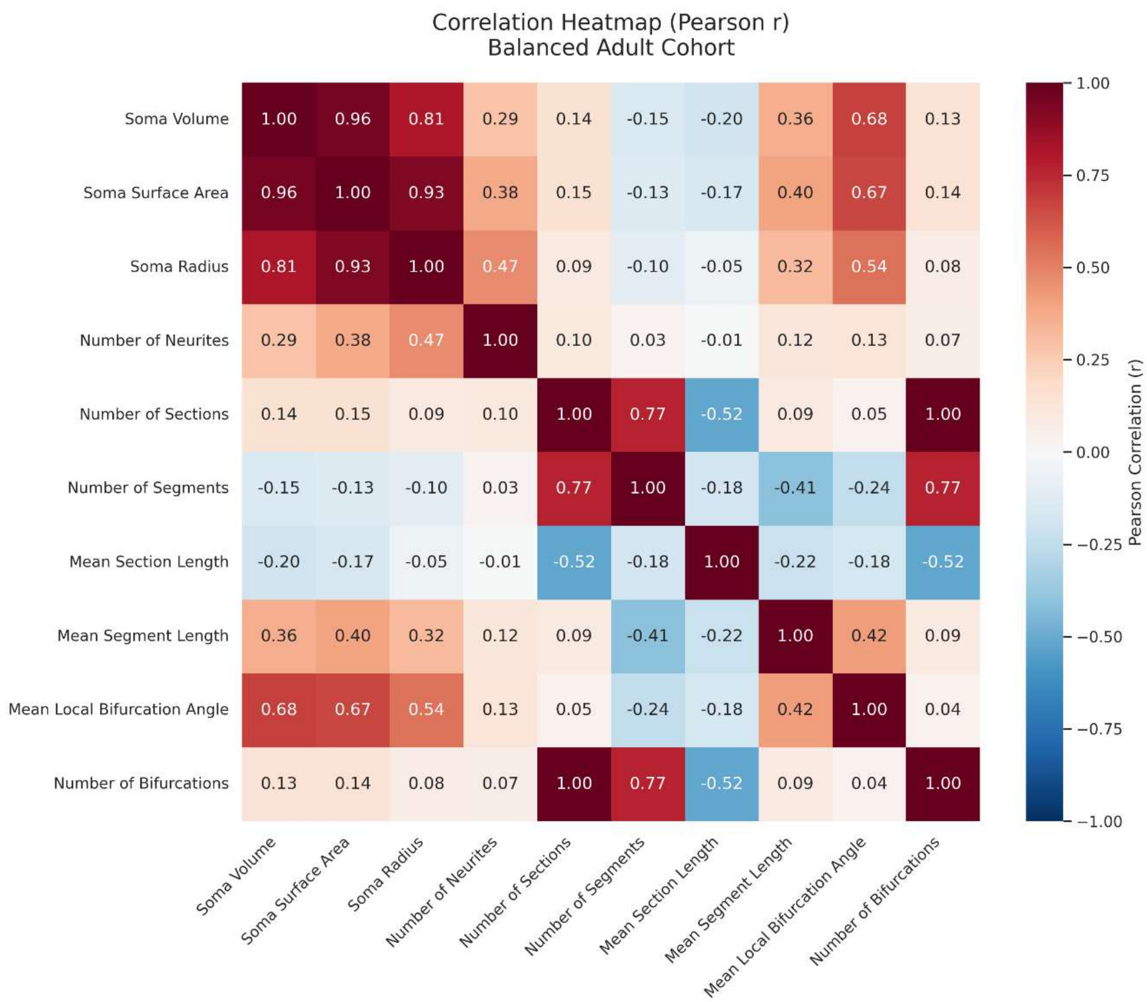


Fig. 3 Heatmap of Pearson correlation coefficients (r) across all features in the filtered, balanced dataset (n = 2,210 cells)

As expected from the geometric dependence between branch points and sections, the number of sections and number of bifurcations were near-perfectly correlated (r = 1.00); these therefore reflect a single axis of branching complexity, and their concordant statistical results should be interpreted accordingly. Soma volume, surface area, and radius intercorrelated strongly (r = 0.81–0.96), showing their geometric interdependence as measures of a roughly spherical cell body. Soma radius showed a moderate positive association with the number of neurites (r = 0.47), suggesting that larger somas support a greater number of primary process projections. This is consistent with the metabolic demands of sustaining extensive microglial arbors (Miao et al. 2023). The negative correlation between mean section length and number of sections (r = -0.52) reflects a trade-off between process reach and branching density. Mean segment length and mean local bifurcation angle correlated positively with soma size metrics (r = 0.32–0.68), indicating that morphologically larger microglia tend to have longer segments and wider branching angles, characteristic of the ramified homeostatic state (Sarlus and Heneka 2017).

3.2. PCA and clustering

PCA was applied to the filtered, class-balanced dataset (n = 2,210 cells) to examine dominant axes of morphological variability. Visualization along the first two principal components is shown in Fig. 4 (a). The two biological groups exhibit an overlapping distribution along both axes; however, the 5xFAD AD cohort occupies a more tightly clustered space at lower PCA1 and PCA2 coordinates compared to the wider-ranging control population. The features contributing most to PC1 loadings were soma surface area, soma volume, and soma radius, while PC2 was primarily influenced by the number of bifurcations, number of sections, and number of segments.

K-means clustering (k = 2) was performed to assess unsupervised structural states within the morphometric data, as seen in Fig. 4 (b–c). When applied across all ten morphometric features, the algorithm produced a silhouette score of 0.611 (Fig. 4 (c)). Restricting the K-means algorithm to the top PCA-informed feature subsets yielded a modestly improved silhouette score of 0.648 (Fig. 4 (b)), suggesting marginally cleaner cluster boundaries when background feature noise is reduced. Crucially, across both configurations the unsupervised boundaries did not segregate the true biological groups. Instead, cluster assignments primarily reflected high-complexity control cells with large PCA1 values versus the densely overlapping core of AD and control cells occupying lower PCA1 space. This pattern is consistent with age-controlled microglial pathology representing a continuous morphological transition rather than a clean binary separation between control and AD cells.

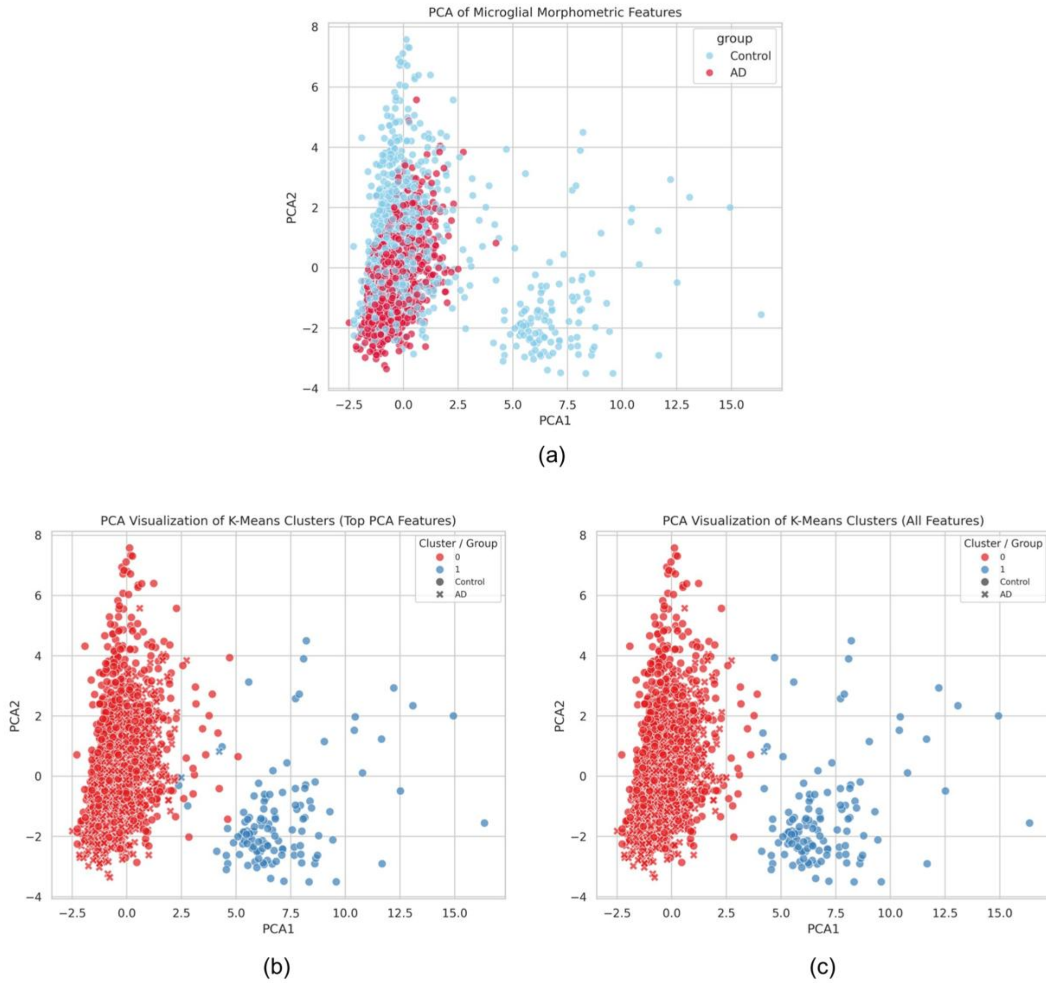


Fig. 4 K-means clustering (k = 2) visualized on the first two principal components of the balanced adult cohort (n = 2,210 cells). (a) Cells colored by biological group (AD vs. control). (b) Clusters from the top PCA-informed feature subset (silhouette = 0.648). (c) Clusters from all ten morphometric features (silhouette = 0.611). Marker shape denotes true group; color denotes assigned cluster

3.3. Supervised machine learning

Supervised classification models were trained on morphometric features from the class-balanced adult cohort to predict disease status. Logistic regression was utilized as a baseline, while Random Forest and Gradient Boosting models were employed to capture nonlinear relationships. Both ensemble methods substantially outperformed the linear baseline, demonstrating exceptional consistency and structural balance between sensitivity and specificity. Specifically, the Random Forest and Gradient Boosting models achieved high classification accuracies of 0.961 ± 0.011 and 0.962 ± 0.005, respectively, whereas the baseline logistic regression model reached only 0.716 ± 0.021. Confusion matrices showed unambiguous separation of 5xFAD and control cells for the ensemble models, exhibiting minimal false positives and false negatives compared to the logistic regression baseline (Fig. 5a–c). In Fig. 5d–f, the receiver operating characteristic (ROC) curves confirm this high discriminative performance across 5-

fold cross-validation. Both the Random Forest (AUC = 0.99) and Gradient Boosting (AUC = 0.99) models demonstrated a marked improvement over the linear model (AUC = 0.78).

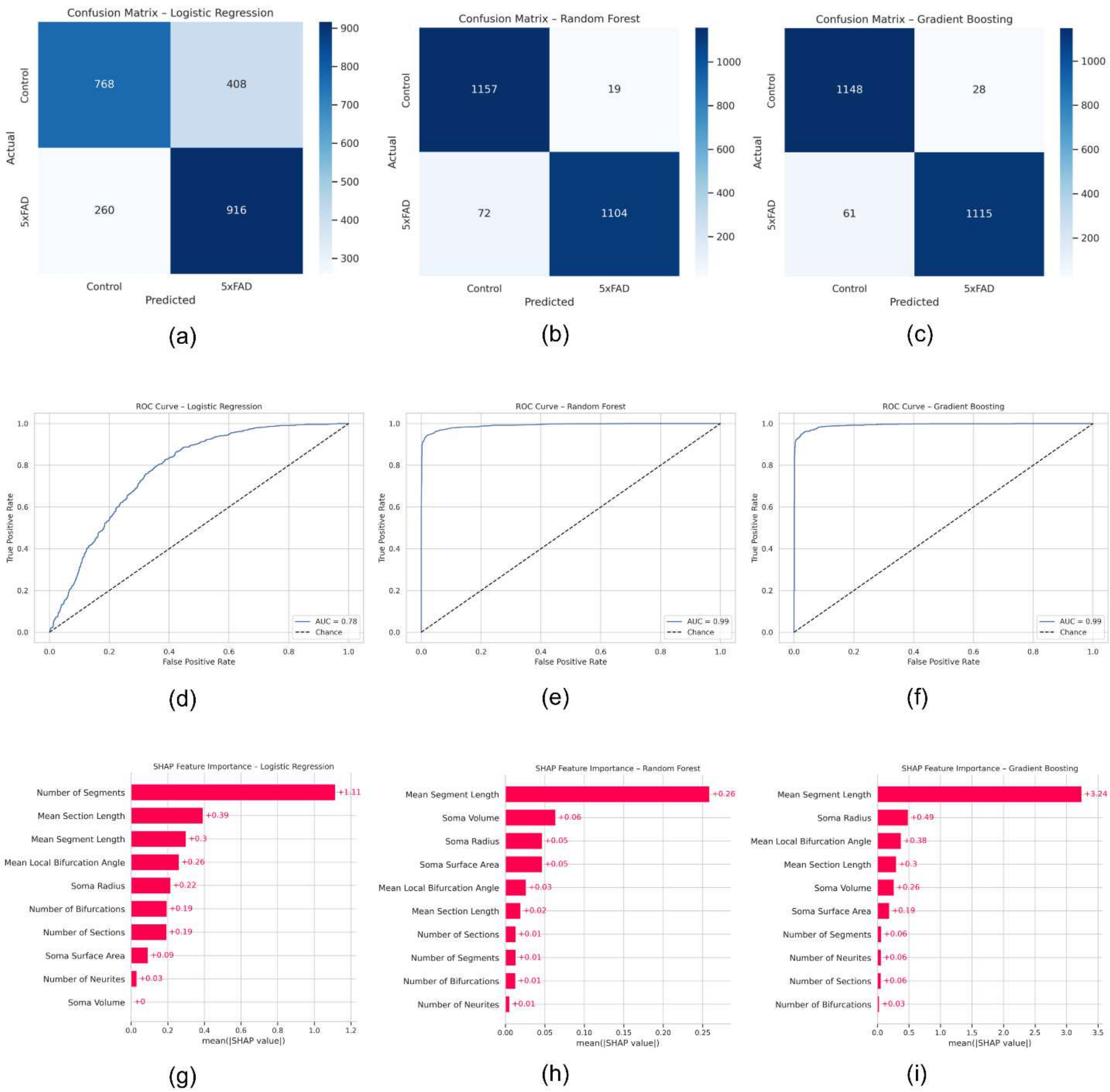


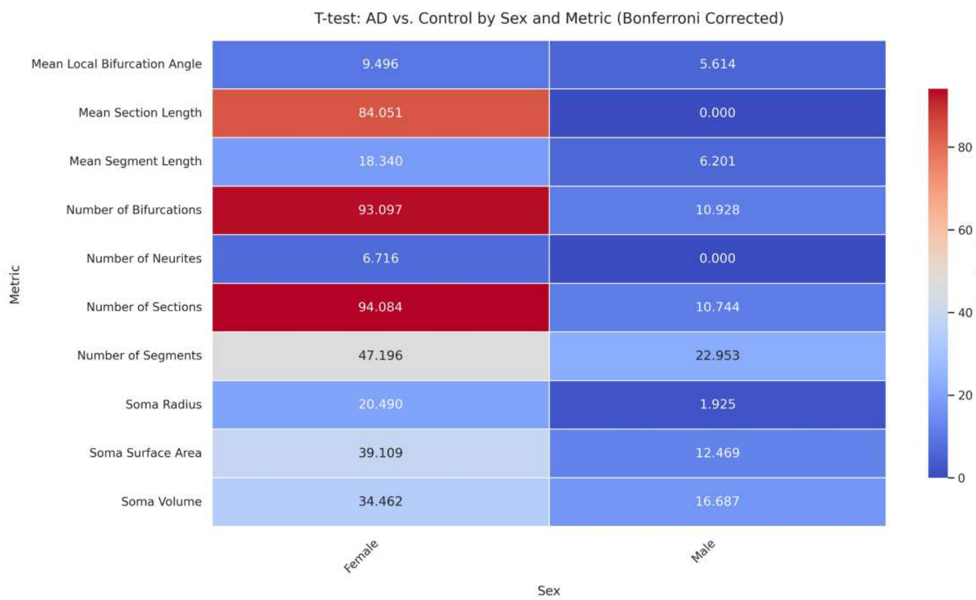
Fig. 5 Comparison of classification performance across three models trained on morphometric data with confusion matrices and ROC curves: (a)/(d) logistic regression, (b)/(e) random forest, and (c)/(f) gradient boosting. (a-c) show confusion matrices and (d-f) show corresponding ROC curves. Feature ranking from SHapley Additive exPlanations (SHAP) illustrates the relative contribution of morphometric features to the (g) logistic regression classifier, (h) random forest classifier, and (i) gradient boosting classifier

SHAP analysis was used to evaluate feature contributions to model predictions (Fig. 5(g–i)). Top predictive metrics varied by model architecture: number of segments ranked highest in Logistic Regression (mean|SHAP value| =

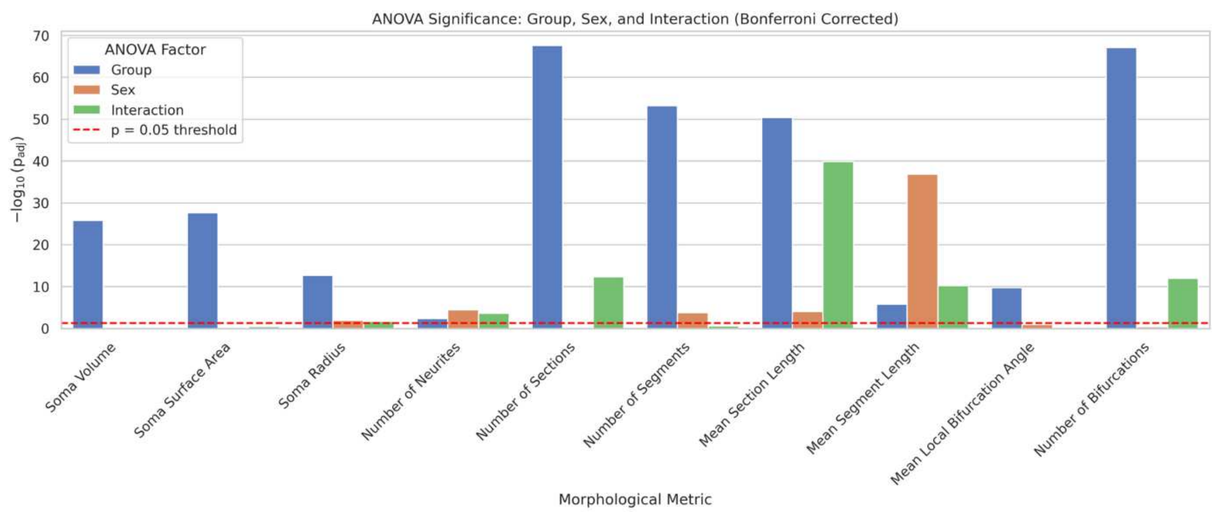
+1.11), while mean segment length was uniquely the most influential feature in both Random Forest (mean|SHAP value| = +0.26) and Gradient Boosting (mean|SHAP value| = +3.24). This shows that alterations in the average length of individual process segments constitute the strongest morphological signal captured by the nonlinear classifiers in the age-matched adult cohort.

3.4. Sex-specific morphological differences (RQ2)

Sex-stratified comparisons were performed using NeuroMorpho.org metadata on the age-matched adult cohort (Fig. 6, Online Resource 7). Female microglia showed larger reductions in the majority of metrics compared to males (Fig. 6 (a)). To control the family-wise error rate across the demographic summary layout, the post-hoc Bonferroni correction adjusted for a family of 20 independent comparisons (10 morphometric features evaluated independently across 2 sexes). Consequently, each raw p-value was multiplied by a correction factor of 20 ($n = 20$). After Bonferroni correction, all metrics remained significant for females ($p_{adj} < 0.05$). For males, most metrics remained significant, with the exception of the number of neurites ($p_{adj} = 1.000$) and mean section length ($p_{adj} = 1.000$). As seen in Fig. 6 (b), group-based stratification accounted for the majority of morphological variance.



(a)



(b)

Fig. 6 Statistical comparison of morphological metrics between AD and control groups, separated by sex

(a) T-test results (Bonferroni-adjusted p-values) comparing AD vs. control groups for each morphological metric across male and female subjects. Warmer colors indicate larger $-\log_{10}$(adjusted p-values), corresponding to stronger group differences

(b) ANOVA results showing the $-\log_{10}$(p-values) significance of the main effects of group, sex, and their interaction for each morphological metric. The dashed line indicates the Bonferroni-corrected significance threshold (adjusted p = 0.05)

T-test analysis stratified by sex revealed larger effect sizes in female microglia across the majority of morphological metrics (panel a), with $-\log_{10}$ (adjusted p-values) consistently higher in females than males for most metrics. Metrics including number of sections (female $-\log_{10}$(adj. p) = 94.084), number of bifurcations (93.097), soma surface area (39.109), and soma volume (34.462) showed particularly pronounced group differences in females, while males exhibited significant but comparatively attenuated effects. Of note, mean section length showed a reversal in direction between sexes: significant and elevated in females (adj. p = 8.828E-85), but non-significant in males after Bonferroni correction (adj. p = 1.000), suggesting a sex-specific reorganization of process architecture that is not simply a uniform scaling of the male response. Two-way ANOVA confirmed disease group as the dominant source of variance for all metrics, with significant group × sex interactions detected for number of sections, number of bifurcations, mean local bifurcation angle, and mean section length (panel b), indicating that the magnitude of AD-associated morphological change differs significantly between sexes and is not simply additive.

Effect sizes across morphometric features varied by sex and disease status (Online Resource 7). Notably, females displayed larger effect size magnitudes across the majority of metrics compared to males. The largest effect sizes were observed for female section number (Cohen's d ≈ −0.89) and female bifurcation count (Cohen's d ≈ −0.88), indicating particularly pronounced AD-associated simplification of the branching arbor in female microglia. Mean section length in females showed a positive effect size (Cohen's d ≈ +0.93), reflecting the paradoxical increase in average section length despite reduced total branching, consistent with partial process retraction and loss of proximal branch points.

3.5. 5CSRTT behavioral analysis (RQ3)

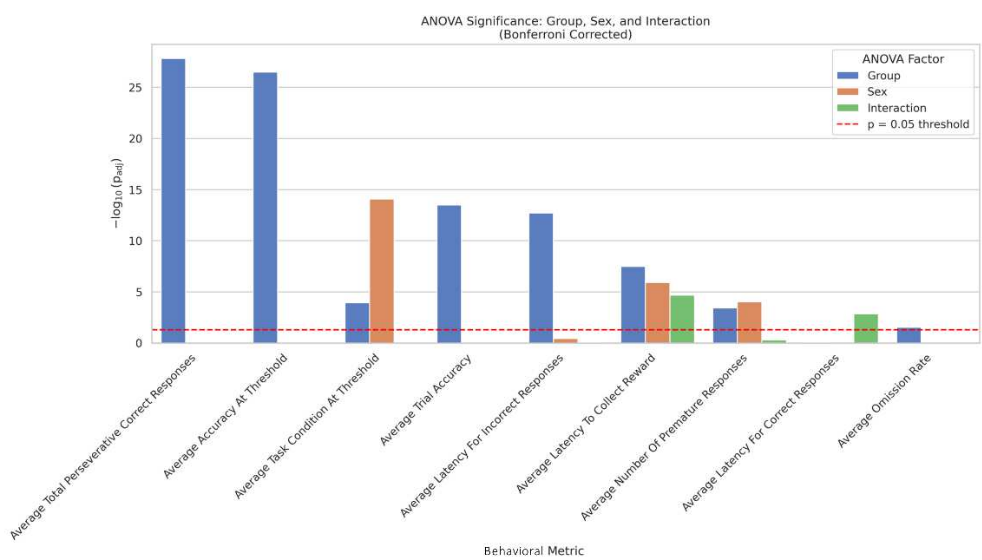
Behavioral data from the 5CSRTT task were analyzed to evaluate whether cellular differences correspond to cognitive outcomes (Figs. 6-7, Online Resource 8). Because behavioral and morphological data derive from separate cohorts, the following comparisons are made at the population level and are not within-subject correlations.

To maintain family-wise error rate control across the behavioral profiling layout, the post-hoc Bonferroni correction adjusted for a family of 18 independent comparisons (9 behavioral domains evaluated independently across 2 sexes). Accordingly, each raw p-value was multiplied by a correction factor of 18 (n = 18). Metrics are reported in full in Online Resource 8 and summarized below.

Males demonstrated significantly stronger and broader cognitive impairments than females across the majority of 5CSRTT metrics (Fig. 7(a), Online Resource 8). Metrics surviving Bonferroni correction in males included average total perseverative correct responses (adjusted p < .001), average accuracy at threshold (adjusted p < .001), average trial accuracy (adjusted p < .001), average latency for correct responses (adjusted p = 2.906E-03), average latency for incorrect responses (adjusted p < .001), average omission rate (adjusted p = 2.585E-02), and average latency to collect reward (adjusted p < .001). Average task condition at threshold approached but did not survive correction in males (adjusted p = 6.990E-02). Females showed a narrower impairment profile: average total perseverative correct responses (adjusted p < .001), average accuracy at threshold (adjusted p < .001), average trial accuracy (adjusted p < .001), average latency for incorrect responses (adjusted p < .001), average number of premature responses (adjusted p = 3.199E-03), and average task condition at threshold (adjusted p = 4.728E-02) survived Bonferroni correction. Notably, average latency for correct responses (adjusted p = 1), average omission rate (adjusted p = 1), and average latency to collect reward (adjusted p = 1) did not survive correction in females and should be interpreted with caution.

As seen in Fig. 7(b), two-way ANOVA indicated that disease status accounted for the largest portion of variance, similar to morphological results.

(a)



(b)

Fig. 7 Statistical comparison of 5CSRTT behavioral performance metrics between AD and control groups, separated by sex

(a) T-test results (Bonferroni-adjusted −log10 p-values) comparing AD vs. control groups for each behavioral metric in female (left) and male (right) subjects. Warmer colors indicate larger −log10(adjusted p-values), corresponding to stronger AD-associated differences. Males show stronger signals across perseverative correct responses, accuracy at threshold, incorrect response latency, and reward collection latency. Females show significant signals primarily for premature responses and incorrect response latency

(b) Two-way ANOVA results showing −log10(adjusted p-values) for the main effects of group (disease status), sex, and their interaction for each 5CSRTT metric. Disease status (group) accounts for the largest portion of variance across most metrics, with a notable sex main effect for task condition at threshold and a significant sex-by-group interaction for average number of premature responses and average latency to collect reward. The dashed line indicates the Bonferroni-corrected significance threshold (adjusted $p = 0.05$)

4. Discussion

This study presents a large-scale neuroinformatics analysis demonstrating that AD is associated with systematic, sex-dependent alterations in microglial morphology and cognitive function. By combining high-dimensional morphometric features with interpretable machine-learning models applied to an age-matched adult cohort, we identify a convergent pattern of structural simplification, including soma- and branch-associated metrics, that distinguishes AD from control cells. Importantly, these sex-stratified effects emerge consistently across thousands of reconstructed cells as well as cognitive assessments drawn from publicly available datasets, supporting the robustness and generalizability of the findings. Rather than identifying a single morphological difference, the results point to coordinated remodeling of microglial architecture consistent with activated or dystrophic phenotypes previously reported in AD (Sarlus and Heneka 2017; Valiukas et al. 2025; Colombo et al. 2022).

4.1. Summary of principal morphological findings

In answer to RQ1, AD microglia in the age-matched adult cohort exhibited reduced branching complexity, lower segment and bifurcation counts, and altered soma-related features relative to controls, with statistically significant group differences observed across multiple metrics, reliably captured by both unsupervised and supervised analytical approaches (Fig. 2). Multivariate analyses further revealed that these features collectively contribute to class separability: principal component analysis showed that variance distinguishing disease and control groups was dominated by soma size and branching topology (Fig. 4), while explainable machine-learning models independently identified similar feature classes as the strongest contributors to classification performance (Fig. 5a–f). The convergence between unsupervised and supervised approaches strengthens confidence that these metrics capture core aspects of AD-associated microglial remodeling rather than model-specific artifacts. Notably, mean segment length emerged as the top predictive feature for both ensemble tree-based classifiers (Fig. 5h, i), suggesting that even as overall process number declines, the geometric properties of surviving processes carry dominant diagnostic information.

A second principal finding is the presence of pronounced sexual dimorphism. Female microglia demonstrated significantly greater morphological simplification than male microglia under AD conditions, with all metrics surviving Bonferroni correction in females while some metrics, notably number of neurites and mean section length, failed to survive correction in males (Fig. 6, Online Resource 7, Bishnoi and Bordt 2025). This indicates that sex differences reflect underlying biological divergence rather than scaling effects or reconstruction bias. Together, these findings establish microglial morphology as a sensitive and quantifiable indicator of AD pathology that varies systematically by sex.

4.2. Morphological alterations and biological interpretation

Microglial morphology is widely used as an indicator of functional state, with highly ramified cells reflecting homeostatic surveillance and simpler morphologies corresponding to activated, dystrophic, or phagocytic phenotypes (Sarlus and Heneka 2017; Valiukas et al. 2025). The reductions in branching and segment number observed here are consistent with prior imaging and morphometric studies reporting loss of structural complexity in microglia proximal to amyloid pathology (Davies et al. 2017; Baron et al. 2014). In this study, the dominant contribution of soma- and branch-associated features to principal components and model explainability metrics (Fig. 4–5) suggests that AD preferentially disrupts processes involved in environmental sensing, synaptic interaction, and spatial coverage.

Interestingly, our SHAP analysis revealed a prominent architectural difference in feature reliance (Fig. 5g–i). While the linear baseline model relied heavily on the absolute count of processes (number of segments; Fig. 5g), the nonlinear ensemble models overwhelmingly prioritized the average length of individual segments (Fig. 5h, i). This indicates that when complex feature spaces and multicollinearity are accounted for, the structural compaction of remaining individual processes carry significantly more robust discriminative information about microglial activation states than global cellular counts alone.

These structural changes may reflect underlying molecular reprogramming. Transcriptomic studies have identified disease-associated microglia (DAM) states characterized by upregulation of inflammatory and phagocytic pathways alongside downregulation of homeostatic genes (Wang 2021; Sobue et al. 2021; Wu et al. 2024). More recently, sex-enriched DAM subtypes have been reported, particularly in female-biased AD models (Wu et al. 2024; Kang et al. 2024; Ocañas et al. 2023). The morphological simplification observed in this study is consistent with these transcriptionally defined states, which supports the interpretation that quantitative morphology shows biologically meaningful shifts in microglial function.

4.3. Sex-specific effects and potential mechanisms

Addressing RQ2, the heightened vulnerability of female microglia observed here, reflected in $-\log_{10}$(adj. p-values) exceeding 90 at the cell level for branching and soma metrics in females compared to single- or low double-digit values in males, aligns with broader literature documenting greater AD prevalence, faster cognitive decline, and increased neuroinflammatory burden in females (Boccalini et al. 2025; Ocañas et al. 2023; Kolahchi et al. 2024; Casaletto et al. 2022; Vila-Castelar et al. 2024). Branching metrics (number of sections and number of bifurcations) showed the largest female-specific effect sizes (Cohen's $d \approx -0.88$ to -0.89; Online Resource 7). Notably, females exhibited a significant increase in mean section length in AD relative to controls (Cohen's $d \approx +0.93$), a metric that did not survive correction in males, suggesting that process elongation as a correlate of branch point loss may be a predominantly female-specific morphological adaptation. Several non-mutually exclusive mechanisms may contribute to this dimorphism, including sex hormone modulation of microglial metabolism and inflammatory signaling, sex-specific epigenetic regulation of immune-related genes, and interactions between estrogen signaling and AD risk alleles such as APOE (Guillot-Sestier et al. 2021; Price et al. 2025; Han et al. 2021; Brown et al. 2008; Guo et al. 2023; Bollinger et al. 2017; Catale et al. 2020).

Importantly, the persistence of significant sex-by-disease interactions after both feature normalization and Bonferroni correction indicates biological divergence in microglial responses to AD pathology, rather than differences in baseline cell size or reconstruction scale. These findings highlight the necessity of explicitly modeling biological sex in computational neuroinformatics studies, particularly when developing biomarkers or predictive models intended for translational use.

4.4. Linking microglial morphology to cognitive performance

In answer to RQ3, sex-specific cognitive profiles identified through 5CSRTT behavioral analysis reveal a pattern in which males exhibit broader and more statistically robust impairments than females.

Sex-specific patterns of microglial simplification corresponded to distinct cognitive deficit profiles at the population level after correction for multiple comparisons (Fig. 6, Online Resource 7, Fig. 7, Online Resource 8). Males exhibited more pronounced and statistically robust deficits across a wider range of 5CSRTT measures, including perseverative correct responses, accuracy metrics, response latencies, omission rate, and reward collection latency, suggesting broader attentional and executive dysfunction. Females showed a narrower but still significant impairment profile, with premature responses and incorrect response latency surviving correction, along with accuracy and perseverative correct responses, consistent with selective disruption of inhibitory control and attentional accuracy rather than generalized attentional failure. This dissociation may reflect the fact that microglial morphological change and behavioral cognitive impairment represent different levels of biological organization with potentially non-linear relationships, or that behavioral compensation mechanisms differ by sex. While this does not directly mirror the morphological finding of greater structural simplification in female microglia, both lines of evidence converge on the conclusion that AD-related alterations are strongly sex-dependent, manifesting differently at the cellular and behavioral levels. Importantly, both sexes exhibited significant AD-associated cognitive impairments after correction, reinforcing the relevance of sex as a biological variable in AD research regardless of the direction of effect.

Microglia play a critical role in synaptic pruning and plasticity, so reductions in branching complexity may limit synaptic surveillance and refinement (Zhao et al. 2024). This provides a plausible cellular pathway linking AD-related neuroinflammation to sex-specific cognitive outcomes, positioning microglial morphology as a potential bridge between cellular pathology and behavior-based cognitive function.

4.5. Methodological considerations and limitations

The use of large, publicly available datasets allows for analyses at a scale rarely achieved in traditional morphometric studies, while transparent preprocessing and interpretable machine-learning methods improve reproducibility and interpretability (Lundberg and Lee 2017; Pedregosa et al. 2011). Age-matching of the adult cohort further strengthens the validity of group comparisons by reducing the confounding influence of developmental or senescent morphological changes on disease-associated signals. Nonetheless, important limitations should be acknowledged. Morphological reconstructions were taken from multiple laboratories employing heterogeneous staining, imaging, and reconstruction protocols, introducing potential sources of technical variability (Fig. 1). Although normalization procedures and large sample size mitigate many such effects, subtle laboratory biases cannot be fully excluded.

A further limitation concerns the unit of analysis. Because individual microglia were treated as independent observations, and multiple cells may originate from the same animal, the effective number of independent biological replicates is lower than the cell counts reported here. NeuroMorpho.org metadata did not permit reliable grouping of cells by individual animal, so animal-level clustering could not be modeled. This likely inflates statistical power and yields very small p-values; reported significance levels should therefore be interpreted as descriptive of cell-population differences rather than animal-level effect estimates. Confirming these findings with hierarchical or mixed-effects models (animal as a random effect) is an important direction for future work.

Additionally, morphological and behavioral datasets were analyzed at the population level rather than within the same experimental subjects, limiting direct correlations between cell morphology and behavior. Finally, while the applied models achieved strong classification performance, feature interactions may be nonlinear and context dependent. More complex modeling frameworks, such as graph-based or spatially informed deep learning approaches, may capture higher-order structural relationships not fully represented by the current feature set (Zhang et al. 2021; Ahmedt-Aristizabal et al. 2021; Thapaliya et al. 2024).

4.6. Implications and future directions

The present findings emphasize the value of quantitative microglial morphology as a scalable and biologically informative marker of AD pathology. The identification of sex-specific morphological signatures reinforces biological sex as a critical variable in both experimental design and computational modeling. Future work integrating morphology with single-cell transcriptomics, spatial proteomics, and longitudinal imaging will enable more direct mapping between structural states and molecular pathways. Extending this framework to human-derived datasets and incorporating hormonal status and genetic background will further clarify mechanisms underlying sex-dependent vulnerability in AD.

5. Conclusion

This study demonstrates that AD induces systematic, sex-dependent alterations in hippocampal microglial morphology that correspond with cognitive impairments. Large-scale morphometric analyses of an age-matched adult cohort and interpretable machine-learning models consistently identified soma- and branch-related features as key indicators of AD pathology, with mean segment length emerging as the strongest predictive feature across tree-based ensemble classifiers. Female microglia exhibited more pronounced structural simplification than males across branching and soma metrics, with a notable sex-specific divergence in mean section length suggesting a female-biased process retraction pattern. Integration with population-level behavioral data revealed that males exhibited broader and more statistically robust cognitive impairments across attentional, accuracy, and reward-related 5CSRTT measures, while females showed a narrower but significant impairment profile concentrated in inhibitory control, accuracy, and error latency metrics. Although the direction of sex differences in behavior did not directly mirror the greater morphological simplification observed in female microglia, both results converge on the conclusion that AD-associated alterations are strongly sex-dependent, manifesting differently across levels of biological organization. Collectively, these findings establish microglial structure as a sensitive, quantifiable marker of AD progression and highlight the importance of incorporating biological sex into neuroinformatics studies and translational modeling. Future work linking morphology with molecular and longitudinal data may further clarify mechanisms contributing to sex-biased vulnerability in AD.

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Data Availability: This study utilized exclusively publicly available, third-party data. Hippocampal microglial morphological reconstructions were obtained from the NeuroMorpho.org repository (https://neuromorpho.org/). Behavioral performance metrics from the 5-choice serial reaction time task (5CSRTT) were obtained from the MouseBytes repository (https://mousebytes.ca/). All analytical code, preprocessing pipelines, and derived datasets generated during this study are publicly available on GitHub at https://github.com/AvniSriram/Sex-Specific-AD-Microglia.git.

Ethics Approval: This study did not involve any new animal experimentation by the authors. All microglial and behavioral data were obtained from pre-existing, publicly archived, de-identified third-party repositories (NeuroMorpho.Org and MouseBytes). Ethical approval for the original generation of these data was obtained by the originating laboratories under their respective institutional animal care and use committees, as detailed in the source publications cited in the Methods section. No ethics approval was required for the secondary analyses presented here.

Supplementary Files

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